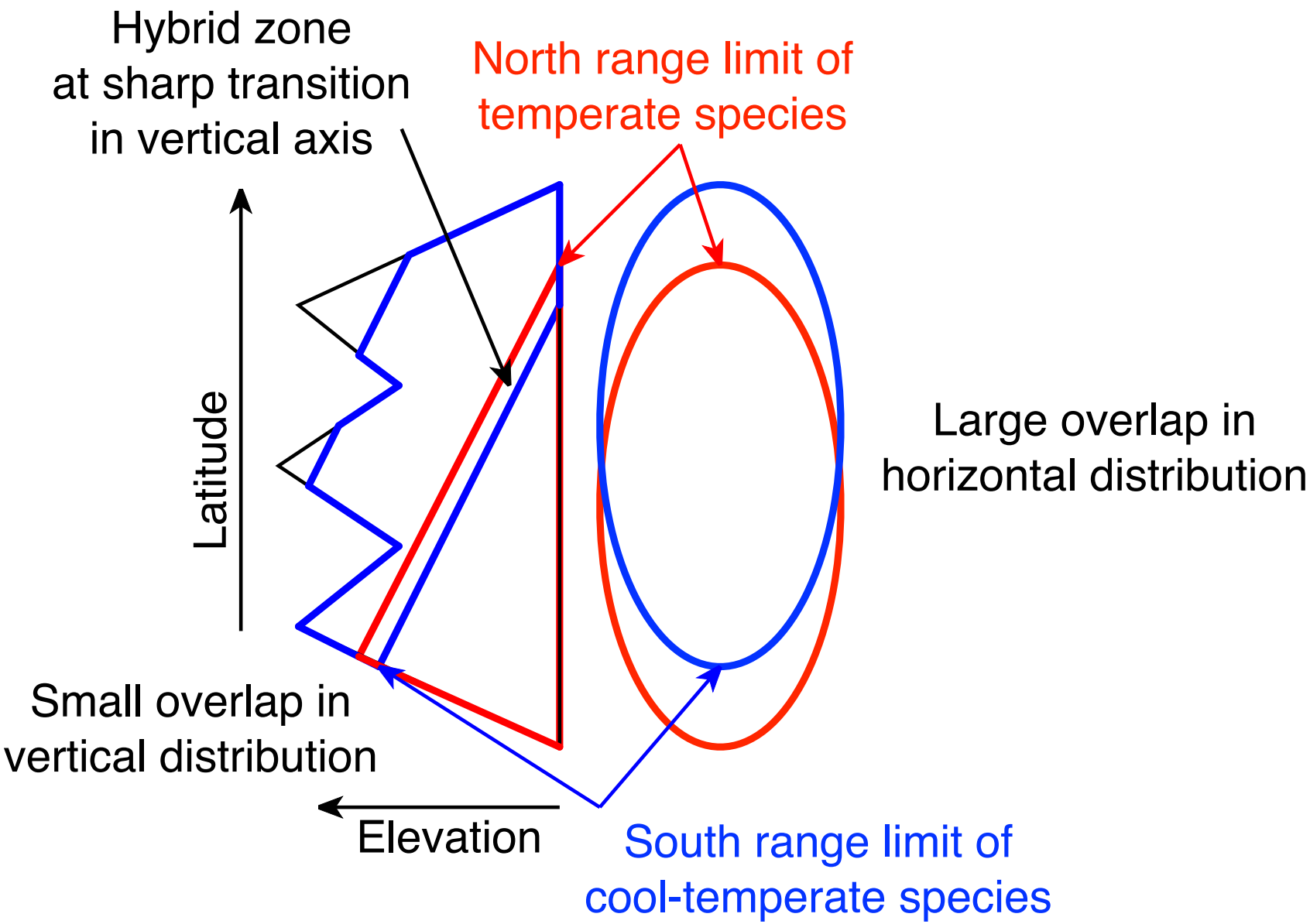
Differential admixture and introgression in *Cerasus* hybrid zones at the south rear edge of cool-temperate *C. sargentii* (オオヤマ) and the north leading edge of temperate *C. leveilleana* (カスミ)

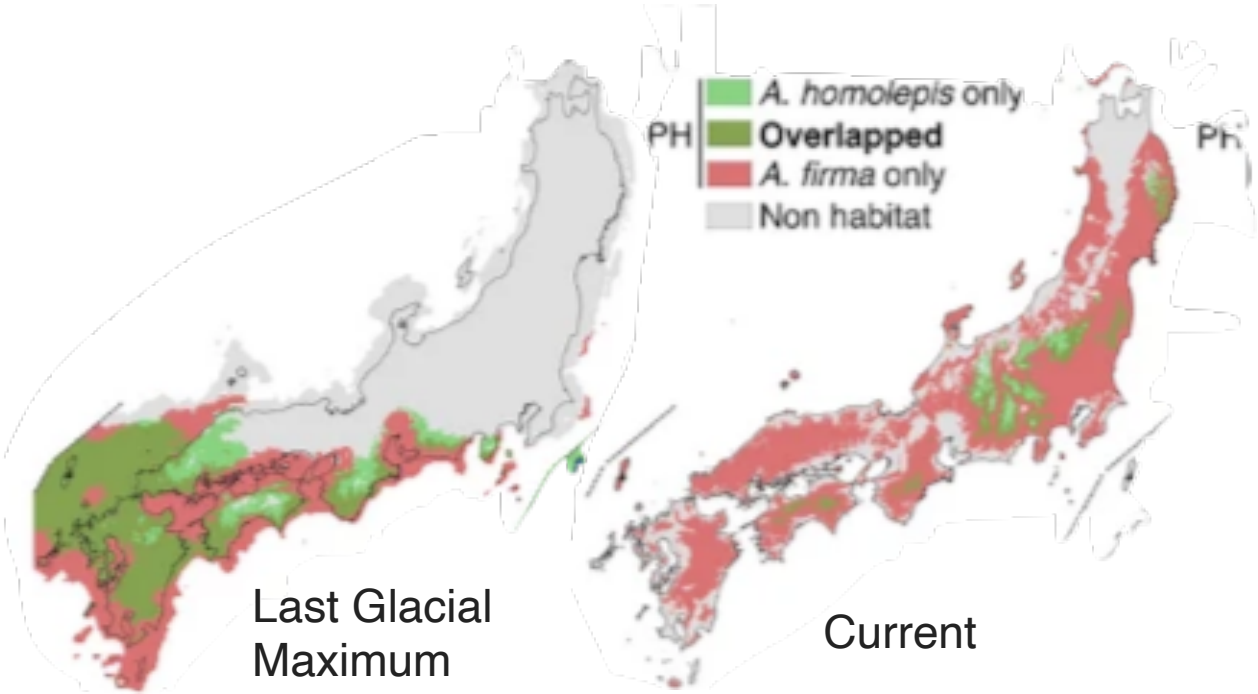
Teruyoshi Nagamitsu (FFPRI)

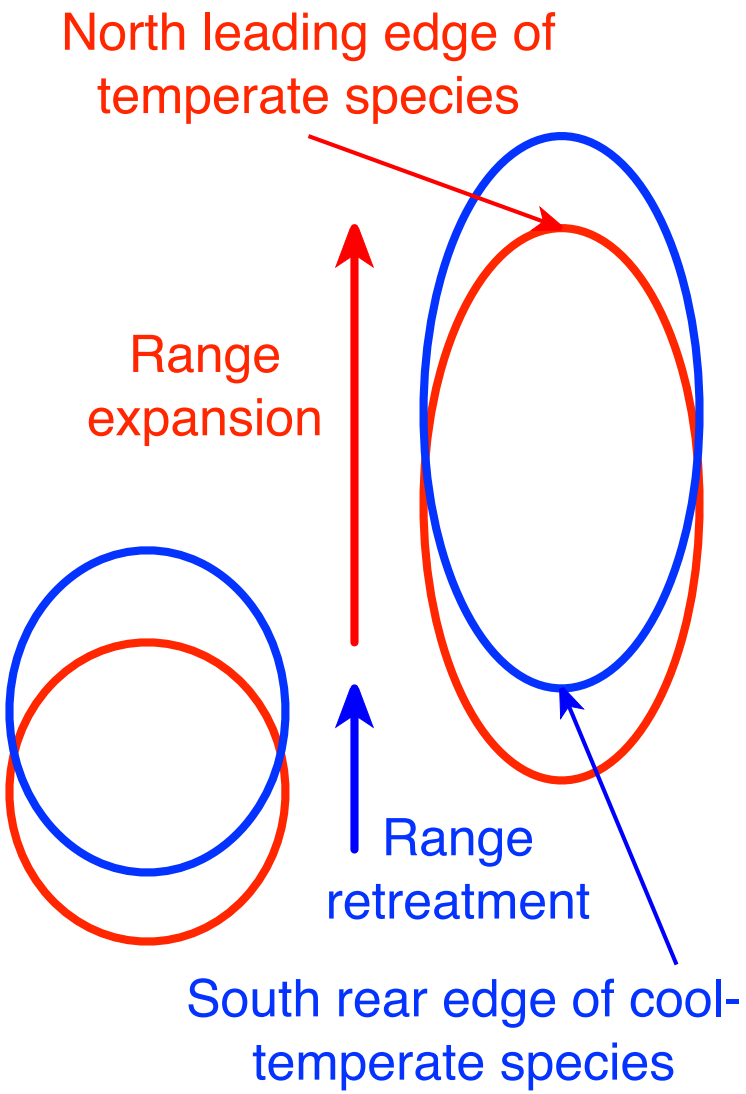
オオヤマ*Cs* (left) has finished flowering, カスミ*Cl* (right) is flowering, their hybrid (center) is finishing flowering (2019 May, Shizunai)

High resolution in vertical axis and low resolution in horizontal axis in environmental gradient on terrestrial landscape



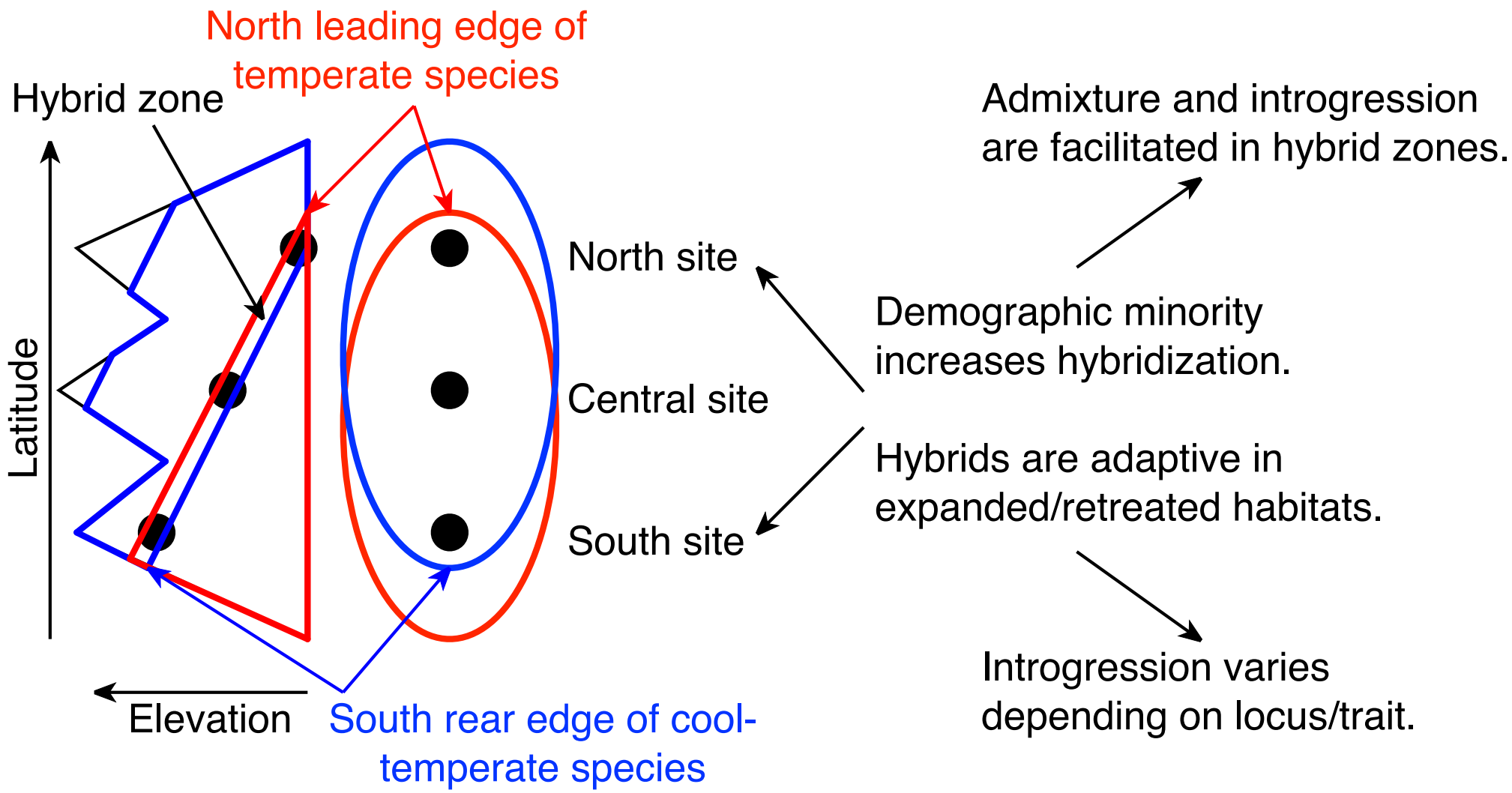
Contrasting demography at north leading and south rear edges in hybrid zones under climate warming trend





Uchiyama, K., Fujii, S., Tsuyama, I. et al. Range shift and introgression of three Japanese Abies species: insights from population genetic data and species distribution modeling. Tree Genetics & Genomes 19, 41 (2023). https://doi.org/10.1007/s11295-023-01617-7

Hypothesis: Different patterns of admixture and introgression in range margins at north leading and south rear edges, in comparison to range core

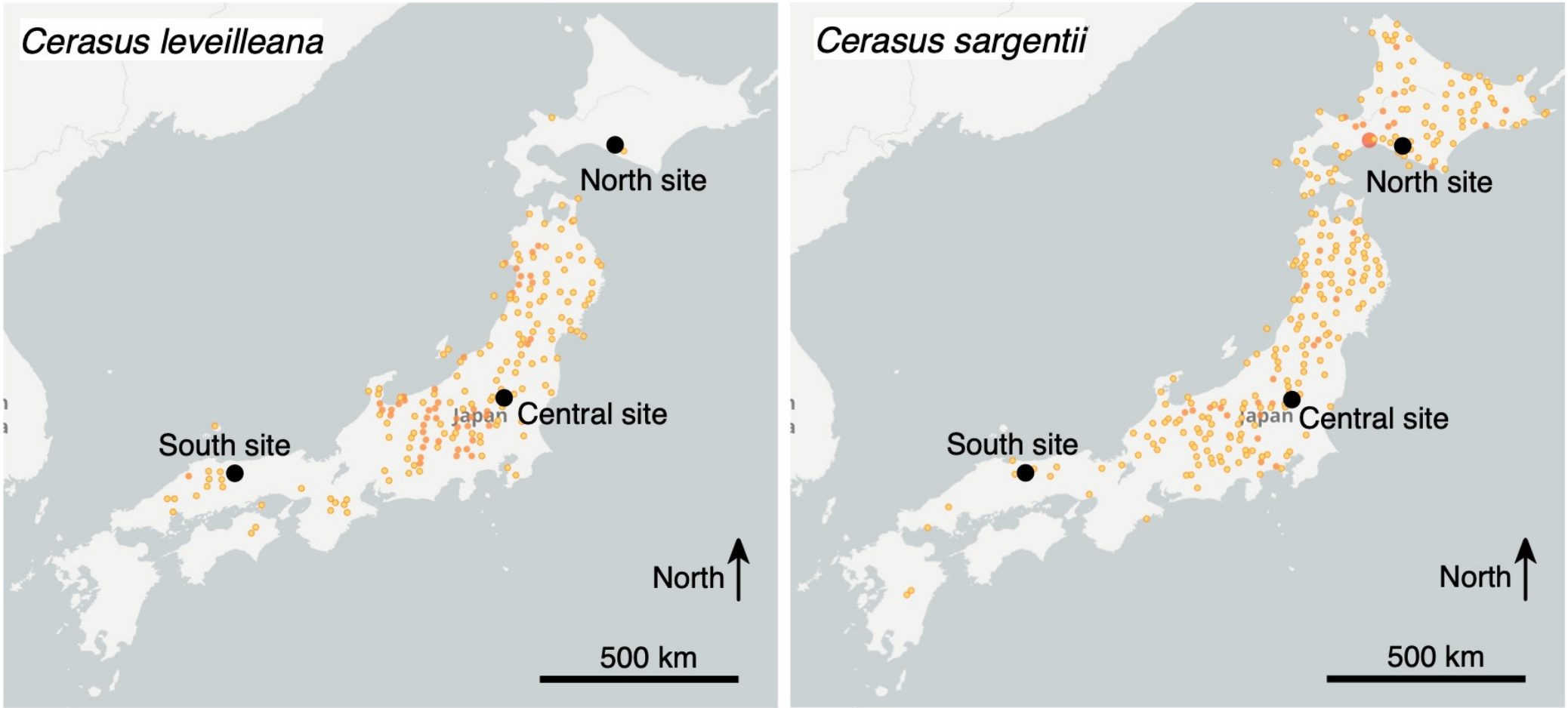


Interfertile *Cerasus* species share pollinators and seed dispersers and form hybrid zones

Temperate species
Cerasus leveilleana

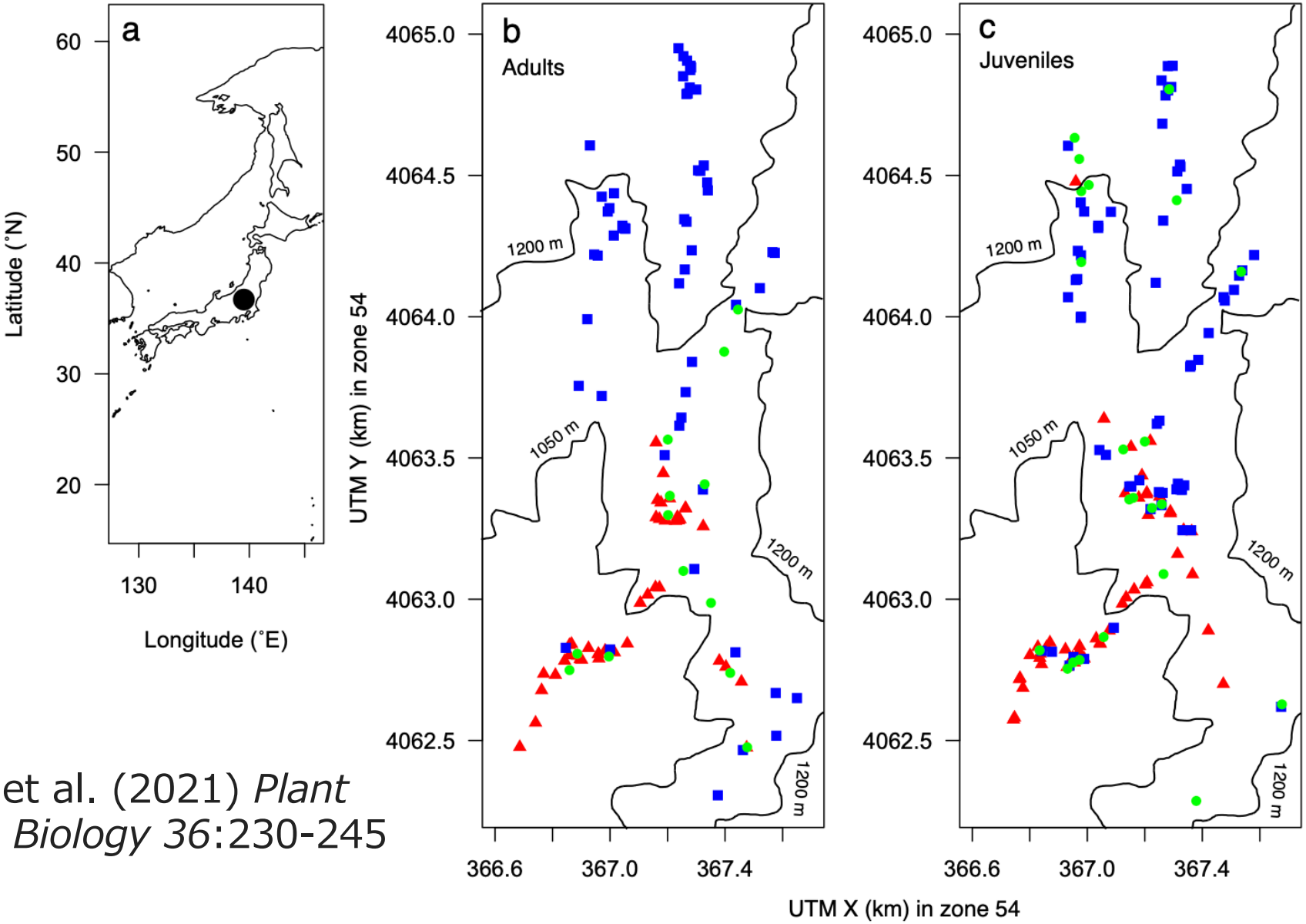
Cool-temperate species
Cerasus sargentii

Geographic distribution in horizontal axis are largely overlapped between *Cl* and *Cs*



Geographic locations of the study sites and occurrence records in Japan obtained from GBIF (Global Biodiversity Information Facility)

Cs is distributed at higher elevations than *Cl* in central site



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Location of central site (a) and spatial distributions of adults (b) and juveniles (c). Red triangles indicate *Cl*, blue squares indicate *Cs*, and green circles indicate their hybrids

Flowering periods are earlier in *Cs* than in *Cl* and intermediate in their hybrid at the same elevation

Hybridization occurs between different elevations

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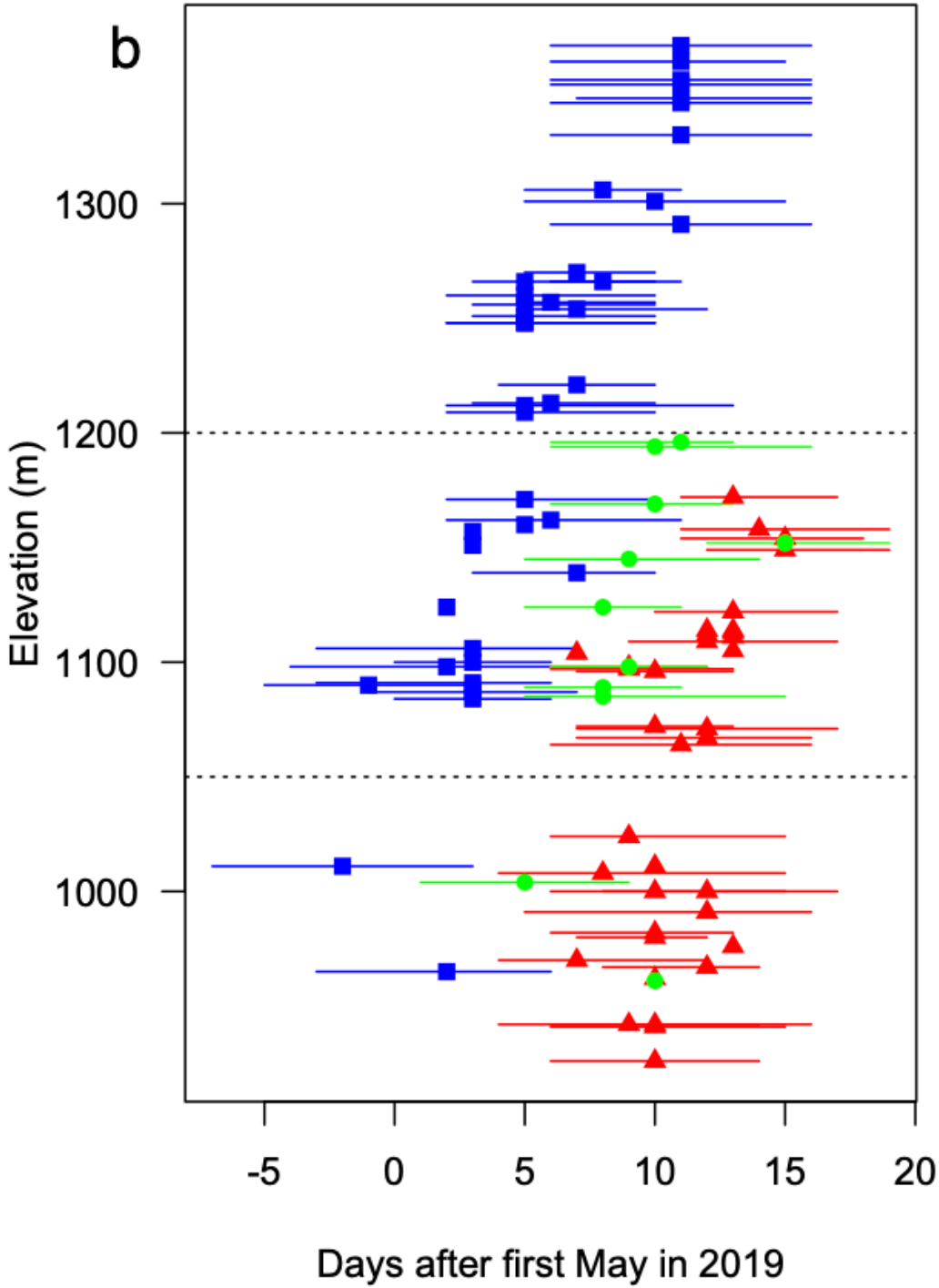
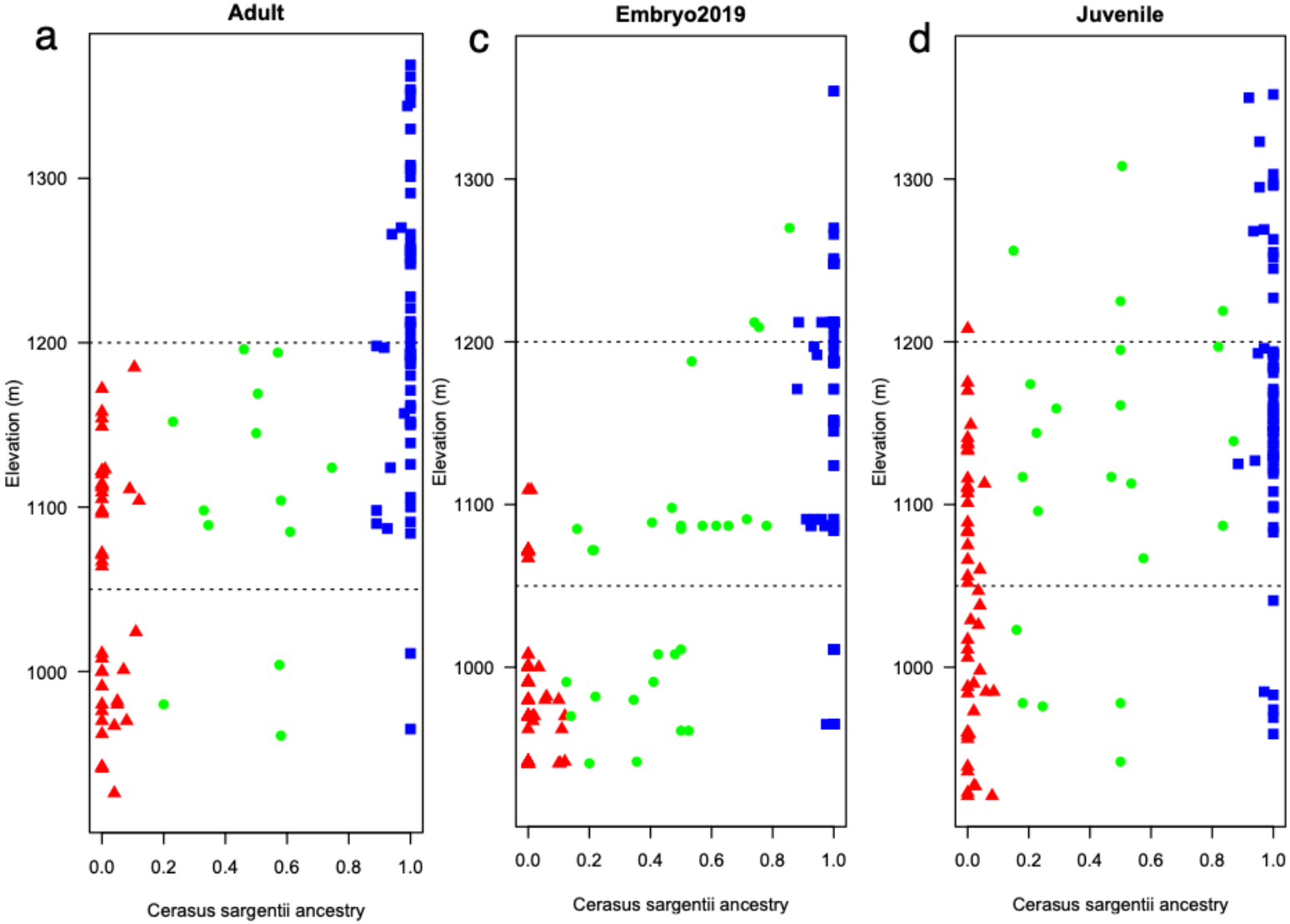


Flowering periods in 2019 along elevational gradient. Red triangles and segments indicate peak flowering date and duration from beginning to end of flowering of *Cl*. Blue squares and segments indicate those of *Cs*, and green circles and segments indicate those of their hybrids.

Recruitment, dispersal, and survival of hybrids along elevational gradient



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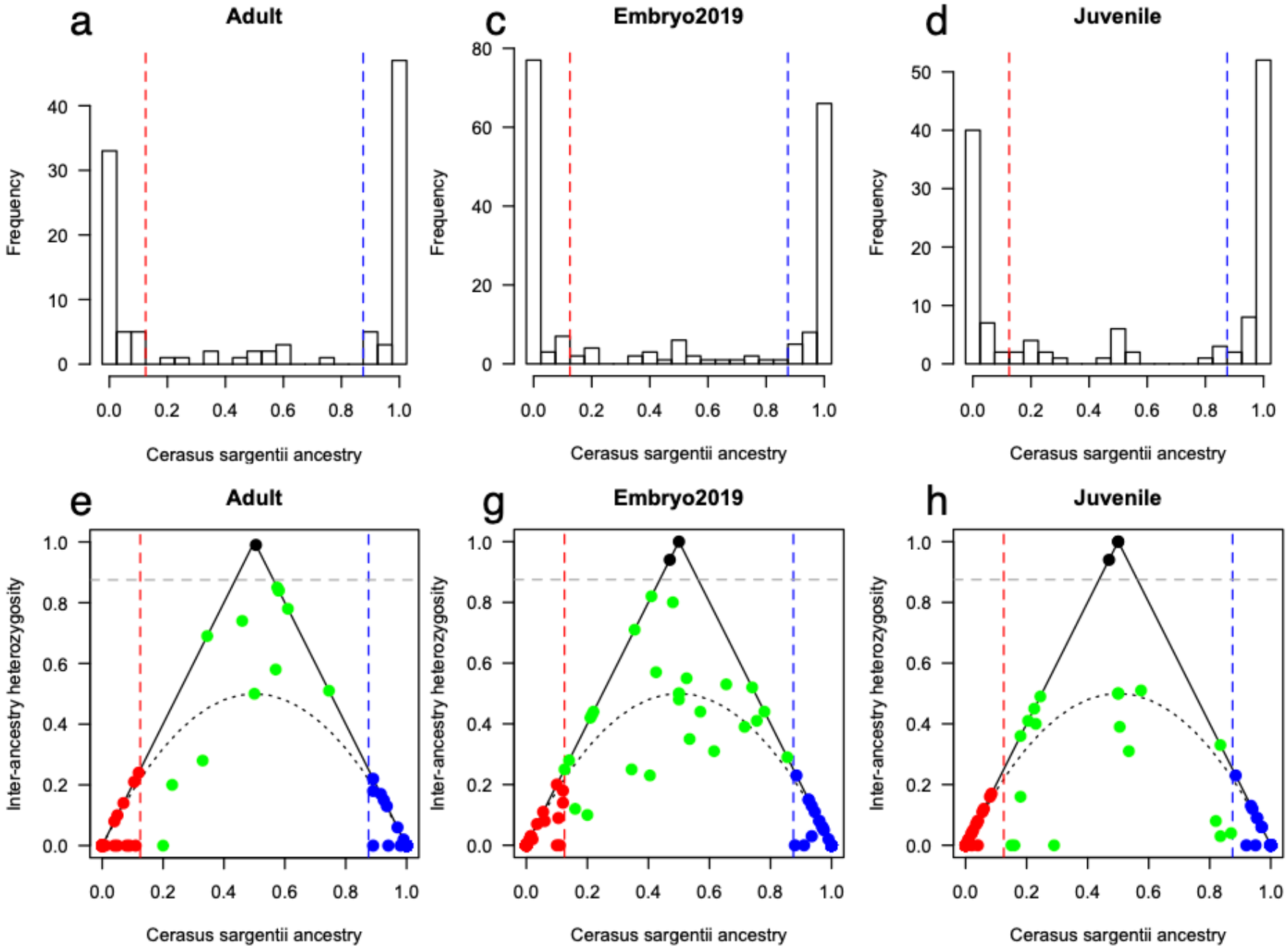
Cs ancestry (hybrid index) of adults (a), embryos in 2019 (c), and juveniles (d) along elevational gradient. Red triangles indicate *Cl*, blue squares indicate *Cs*, and green circles indicate their hybrids.

F1 hybrids are rare, and introgressive hybridization occurs



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Hybrids are recruited but backcrossed to *Cl* and *Cs* and removed in survival process

Mother		Embryo				
Taxon	Number	*C. leveilleana*	Hybrid		*C. sargentii*	Total
C. leveilleana	30	203	23	10.2%	0	226
Hybrid	6	23.7% 9	24		5	13.2% 38
C. sargentii	29	0	12	11.1%	96	108
Total	65	212	59	15.9%	101	372
		C. leveilleana	Hybrid		*C. sargentii*	Total
Juvenile		49	22	16.5%	62	133
Adult		43	13	11.7%	55	111

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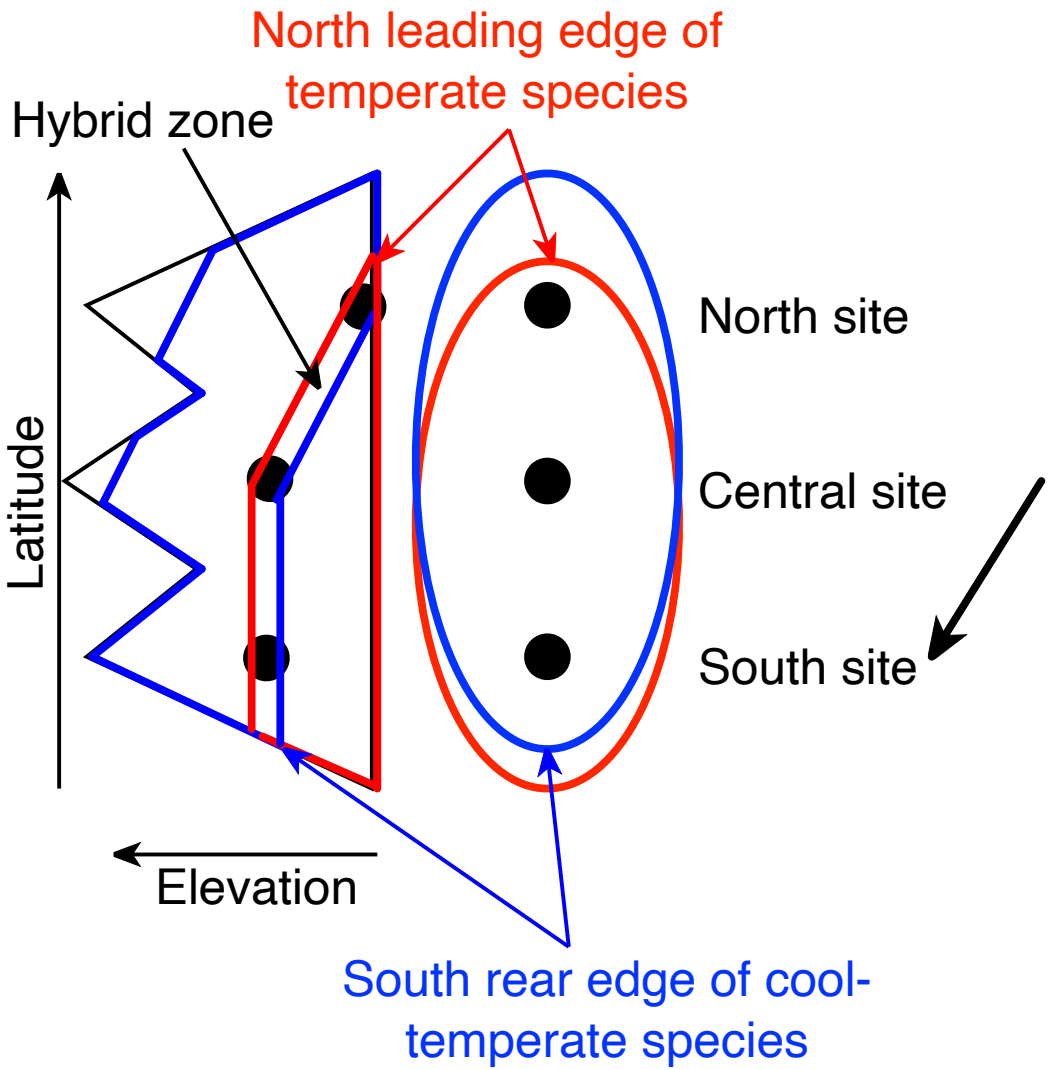
Results: Differential admixture and introgression are found in hybrid zone at only south rear edge, where climate conditions are warmer, in comparison to range core



1) Warmer climate conditions

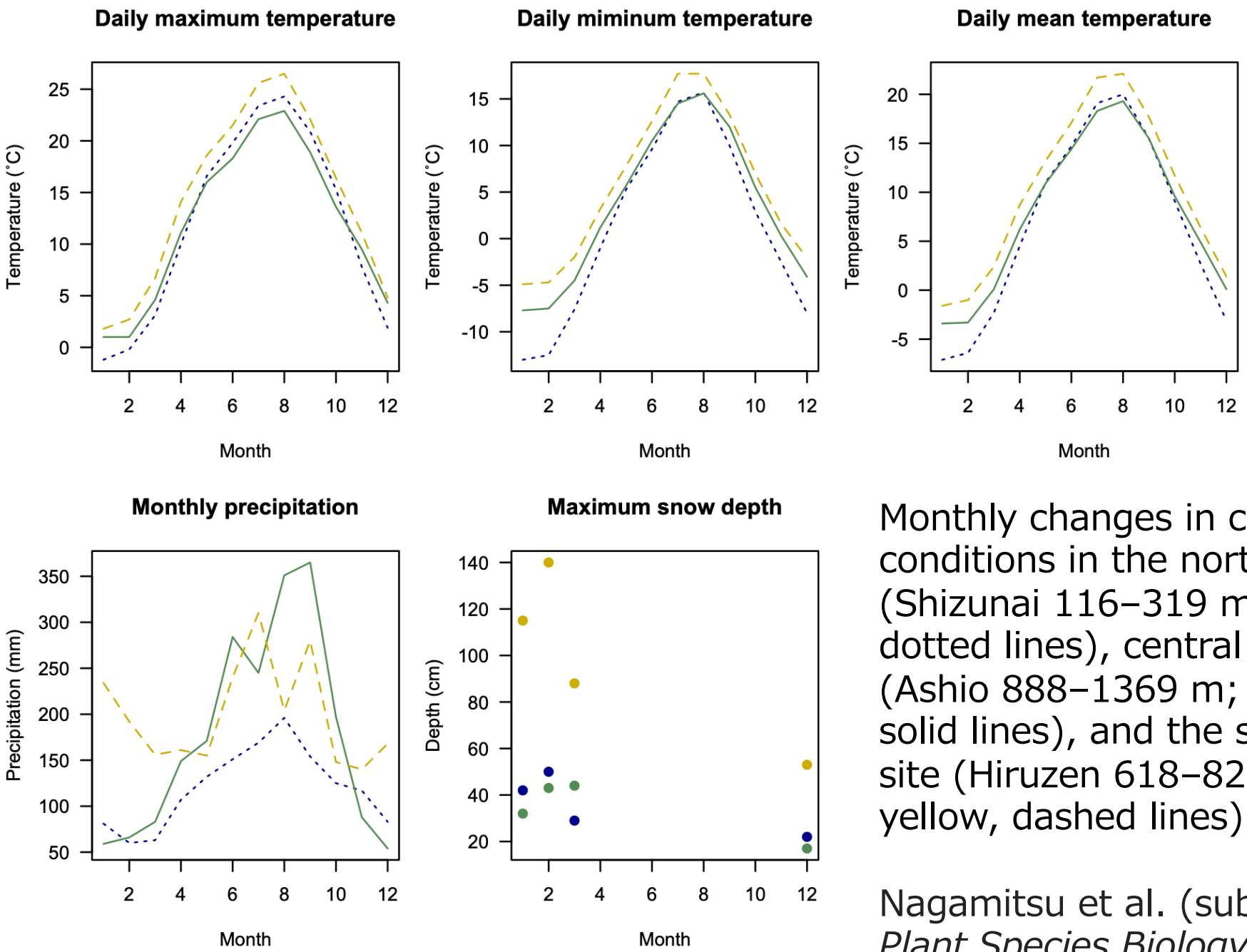
2) More intensive admixture and introgression

3) Gentler cline of genetic admixture in juveniles on elevation

4) Gentler cline of phenotypic introgression in leaf trichome density on genetic ancestry

5) Steeper cline of phenotypic introgression in flowering periods on genetic ancestry

Climate conditions are warmer in south site



Monthly changes in climate conditions in the north site (Shizunai 116–319 m; blue, dotted lines), central site (Ashio 888–1369 m; green, solid lines), and the south site (Hiruzen 618–828 m; yellow, dashed lines).

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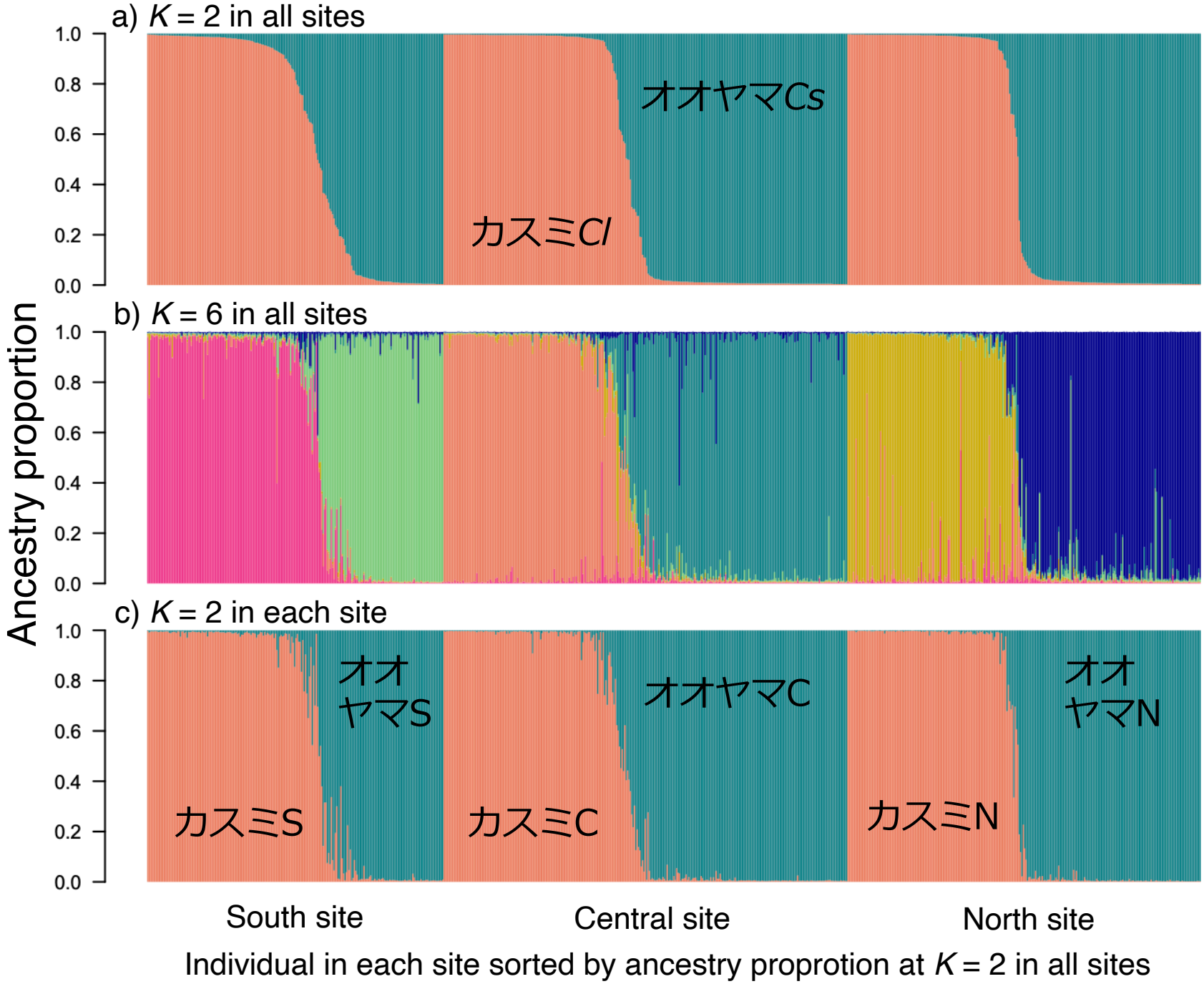
More intensive admixture in south site



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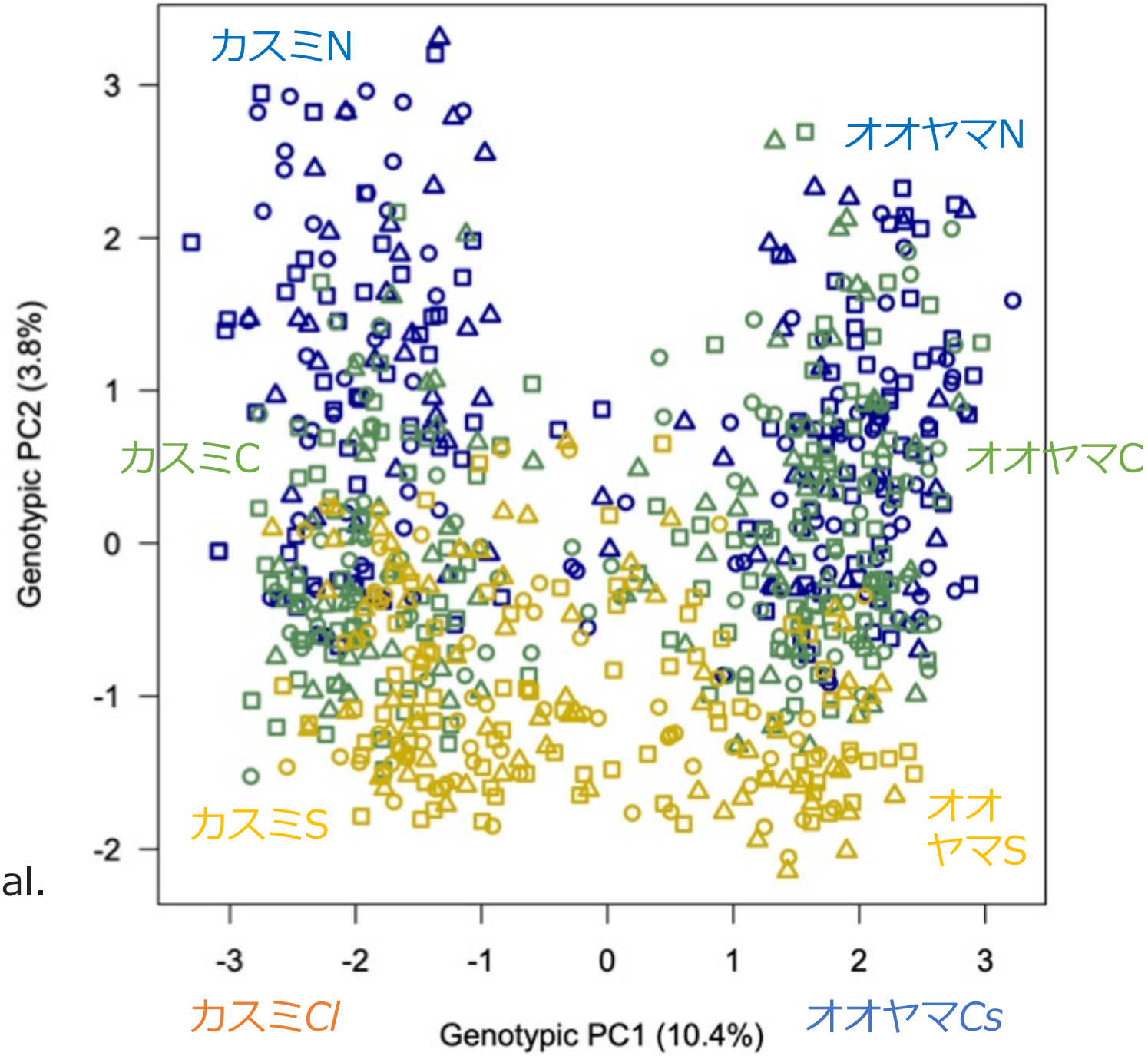
More intensive admixture in south site



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Lower genetic differentiation between local *Cl* and *Cs* populations in south site, suggesting more introgression

Genetic diversity of six local populations of *Cl* and *Cs* selected by *Cs* ancestry. Observed and expected heterozygosity (*HO* and *HE*) and inbreeding coefficient (*FIS*) of each population and pairwise genetic differentiation (*FST*) between populations estimated using ENA method are shown.

Population selected by *Cs* ancestry	H_O	H_E	F_{IS}	Pairwise F_{ST} estimated using ENA					
				Cs N	*Cs* C	*Cs* S	*Cl* N	*Cl* C	*Cl* S
C. sargentii in site N (*Cs* N)	0.648	0.674	0.040	0.000	0.056	0.063	0.148	0.146	0.124
C. sargentii in site C (*Cs* C)	0.635	0.697	0.057		0.000	0.065	0.142	0.142	0.127
C. sargentii in site S (*Cs* S)	0.671	0.692	0.030			0.000	0.150	0.130	0.104
C. leveilleana in site N (*Cl* N)	0.701	0.759	0.076				0.000	0.055	0.067
C. leveilleana in site C (*Cl* C)	0.723	0.767	0.090					0.000	0.062
C. leveilleana in site S (*Cl* S)	0.773	0.781	0.010						0.000

オオヤマ種内地域間

カスミ種内地域間

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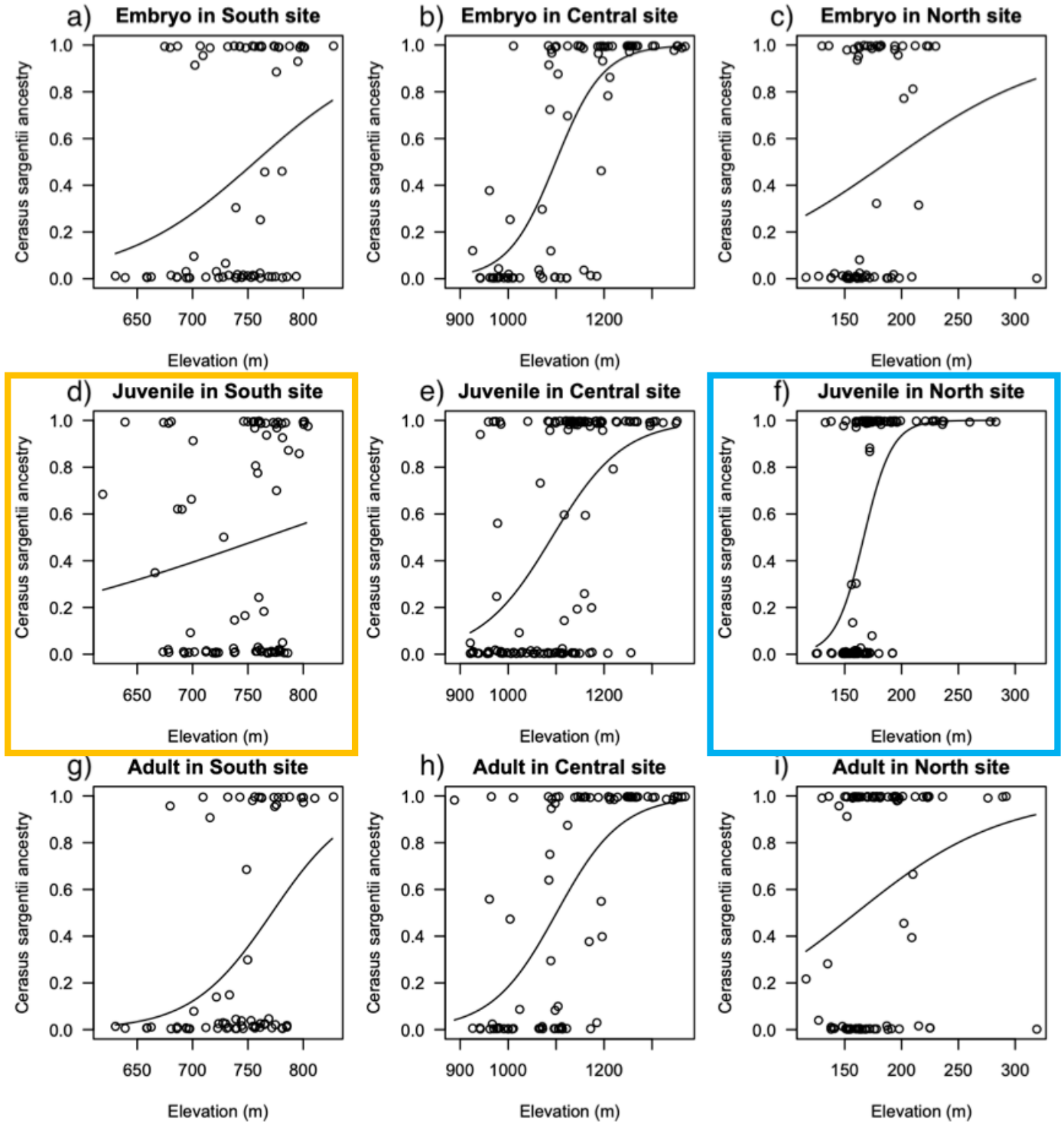
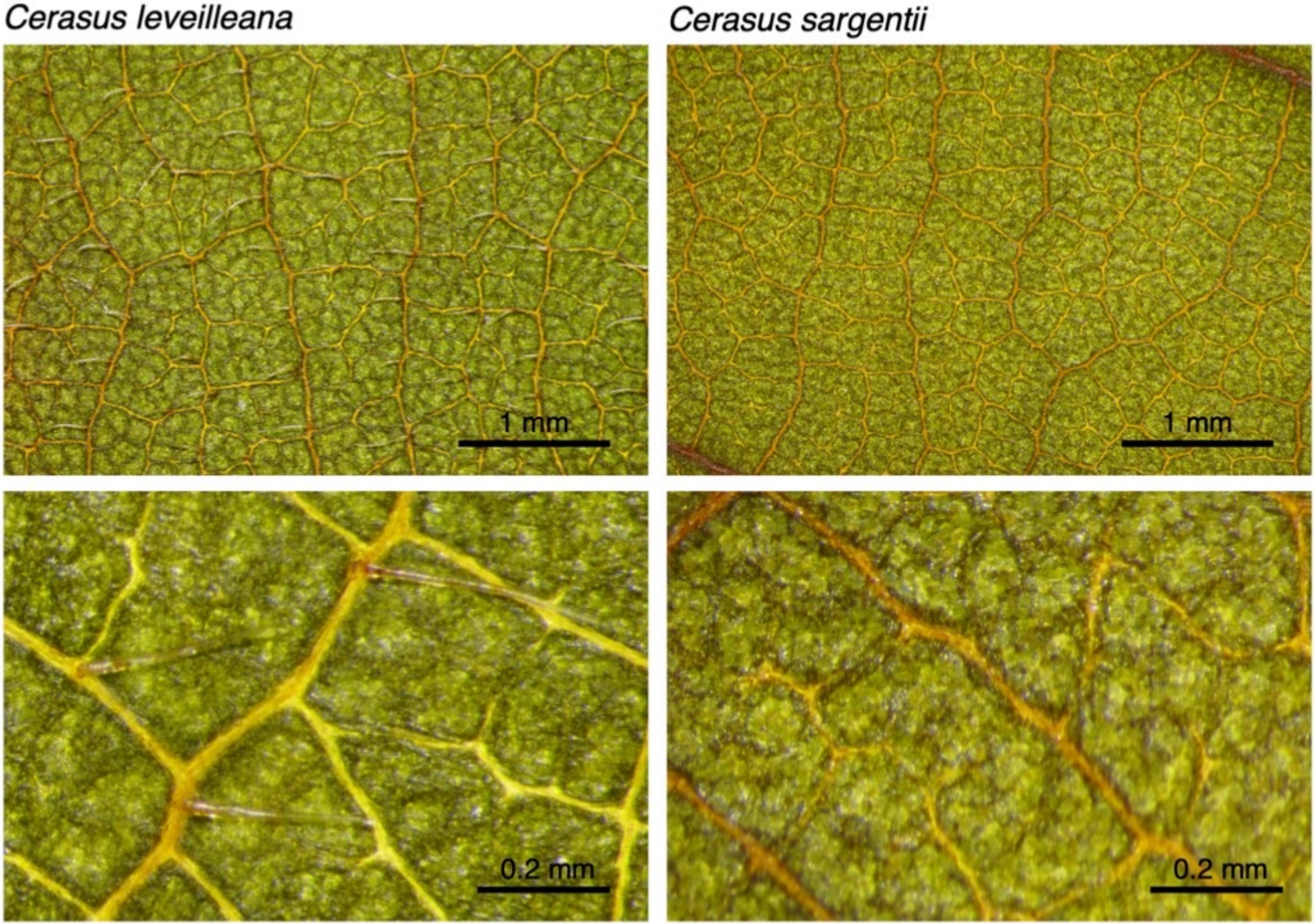
Gentler cline of genetic admixture in juveniles on elevation in south site:
Accumulation of dispersed hybrid seedlings and their removal until adult

Steeper cline in juveniles in north site:
Species-specific habitat preference in regeneration and their competition until adult



Leaf hairs (trichomes) are present in *Cl* but absent in *Cs*



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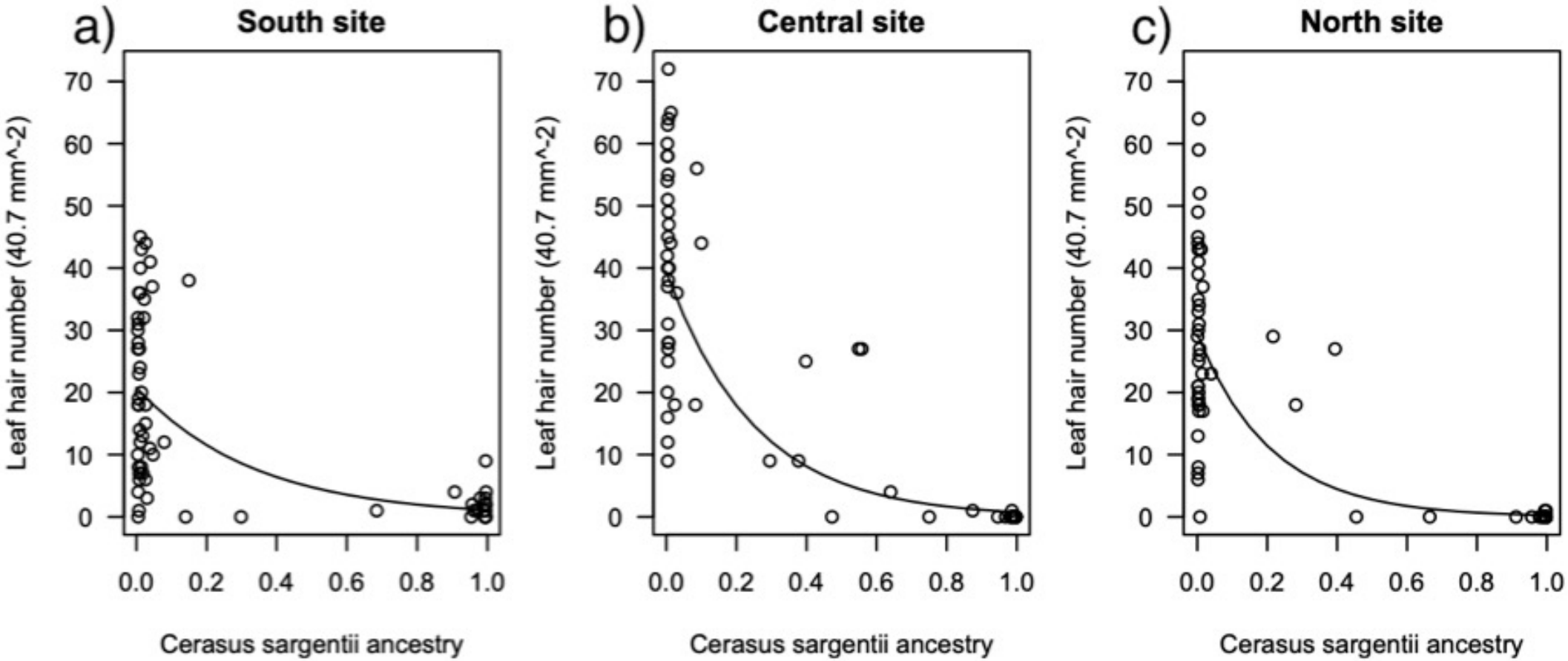
Gentler cline of phenotypic introgression in leaf trichome density on genetic ancestry in south site, suggesting facilitated introgression



Trichome density decreased as elevation increased in each site, suggesting that abiotic and biotic stresses changing with elevations are related to this trait.

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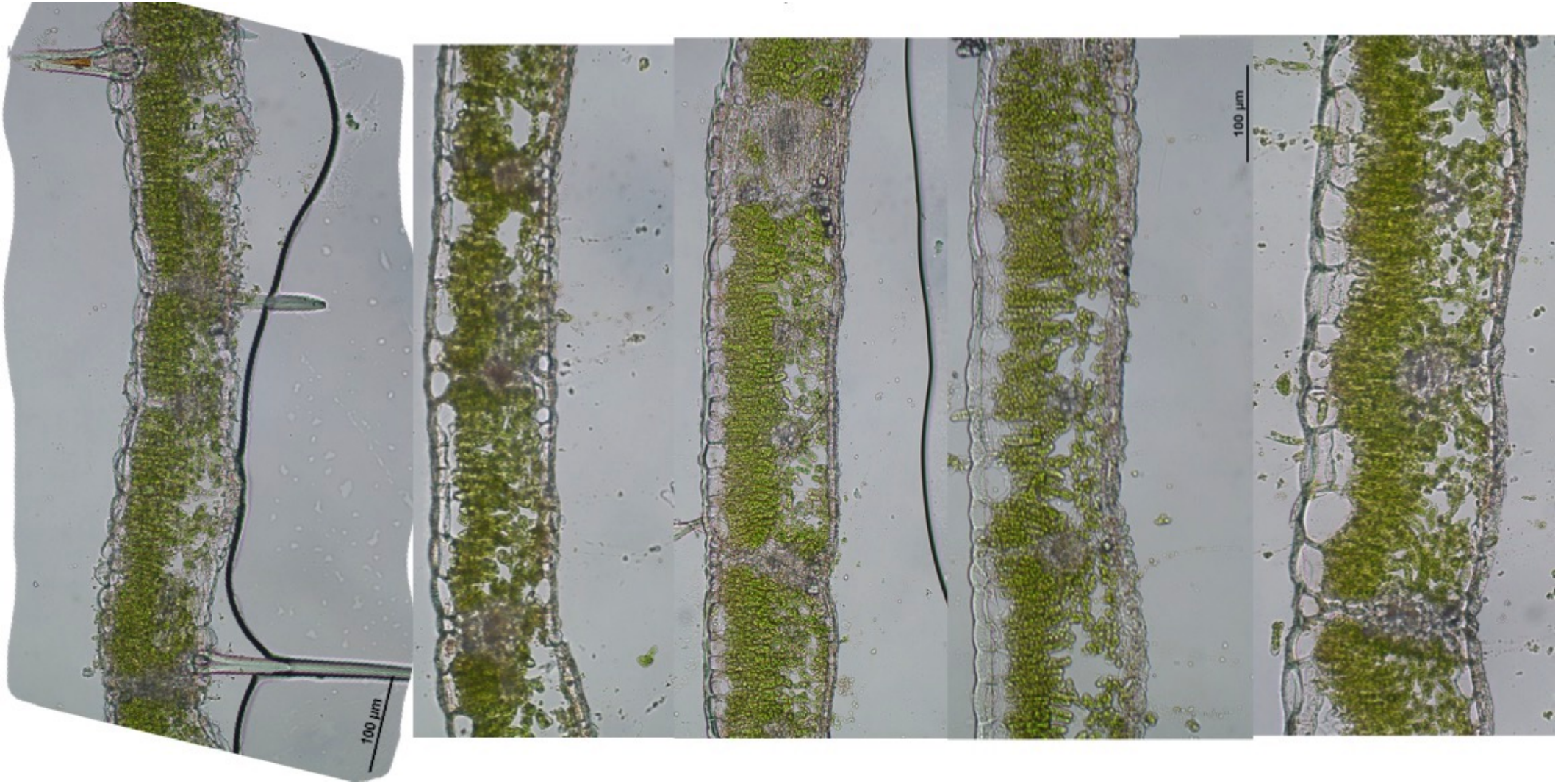
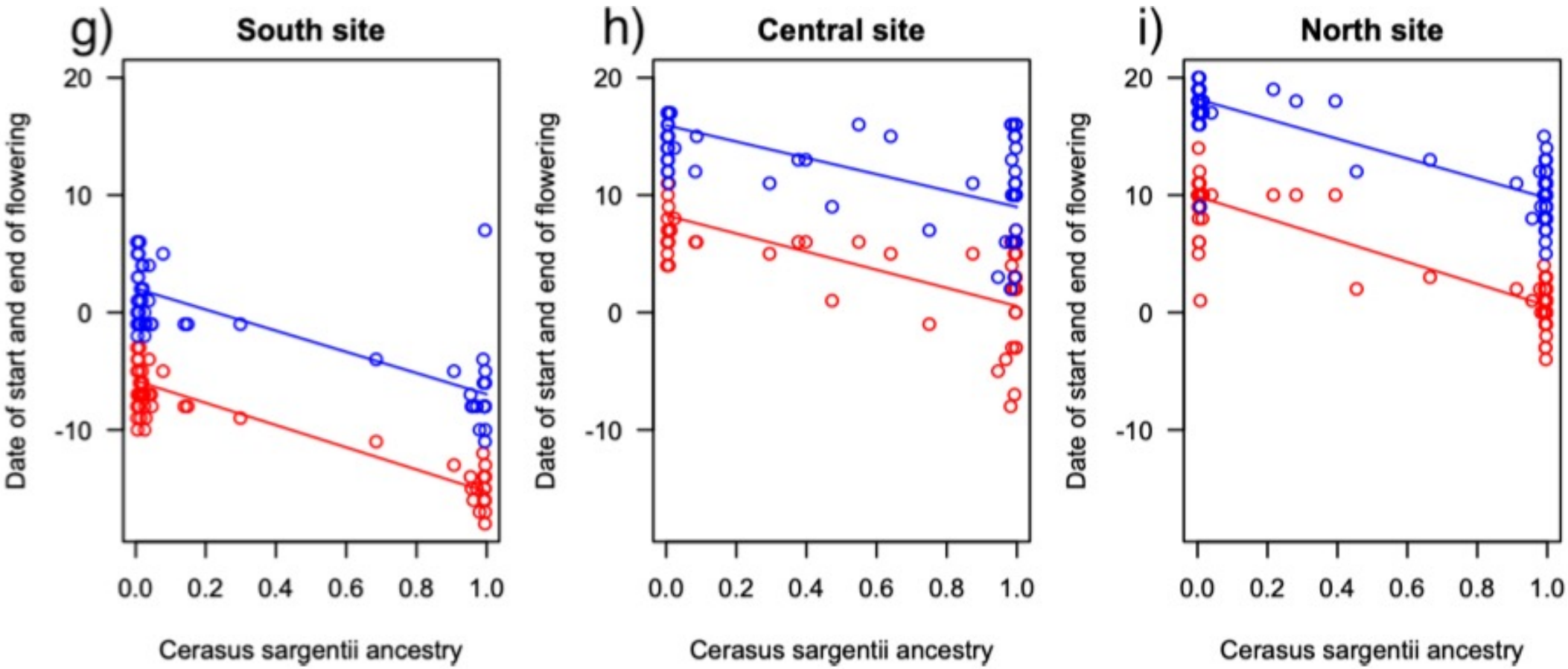
Leaf hairs (trichomes) and thick epidermis are two contrasting strategies for leaf protection in *Cerasus*



チョウジ　マメ　エドヒガン　オオヤマ　オオシマ

Steeper cline of phenotypic introgression in flowering periods on genetic ancestry in south (& north) sites, suggesting impeded introgression and maintenance of premating reproductive barrier



Start (red) and end (blue) of flowering of *Cl* and *Cs* had little overlap at the same elevation in each site.

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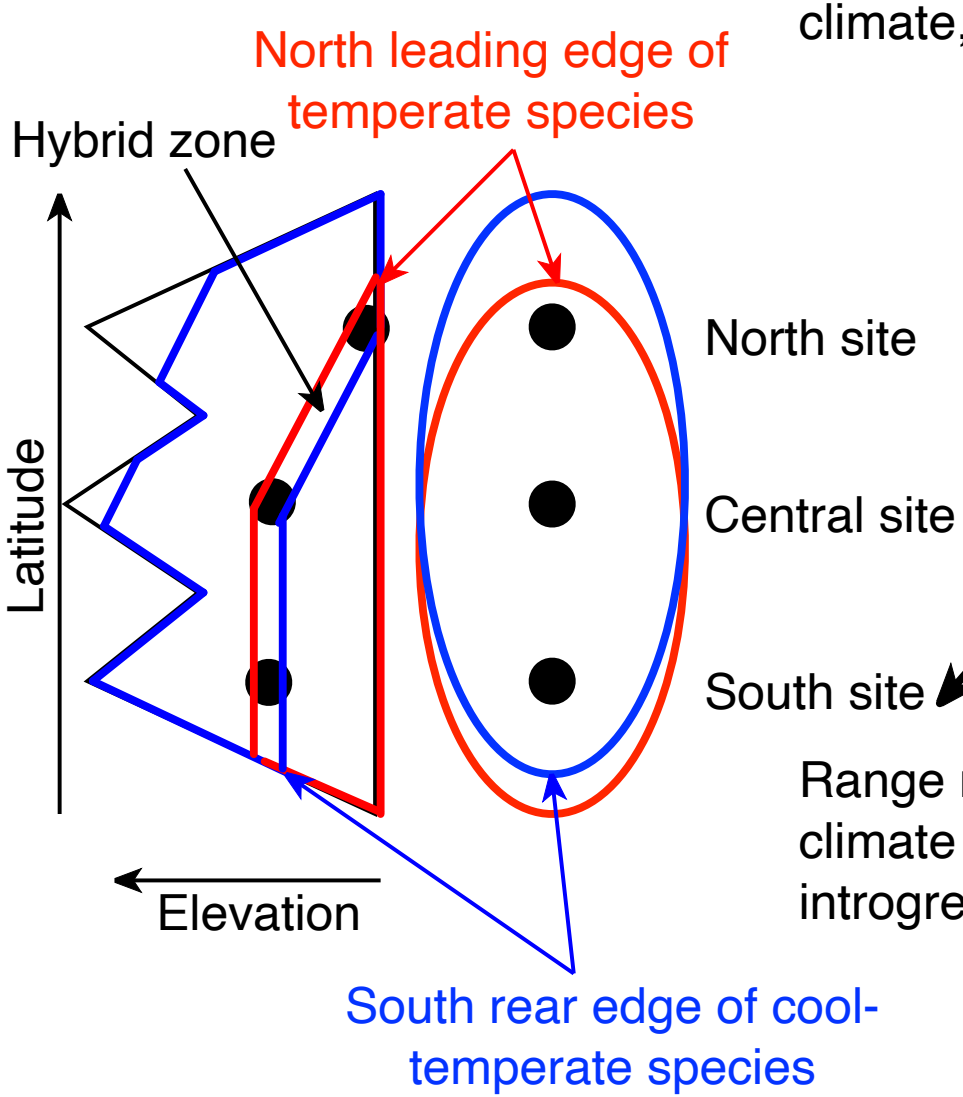
Perspectives: Adaptive introgression of *Cl* alleles to *Cs* genomes at loci associated with adaptation to warmer climate



Hybrid zone has moved to lower elevation in warmer climate, suggesting asymmetric introgression

Admixture and introgression are facilitated at south rear edge.

Demographic minority increases hybridization.

Hybrids are adaptive in retreated habitats.

Range retreatment in climate change affects introgressive hybridization

Phenotypic introgression are facilicated in leaf trait of stress tolerance and impeded in flowering trait of pre-mating reproductive barrier